# Henry Hsu

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## SUMMARY

**Backend engineer who owned the API, data pipeline, and test infrastructure carrying a B2B platform from 3K to 20K users and enterprise rollouts at CTBC Bank, MSI, and HPE. Optimized 140+ APIs (27.7% faster), scaled async pipelines from 3M to 20M+ text chunks, and cut production hotfixes by 90%; built integration test infrastructure at Google. M.C.S. at Rice, Dec. 2026.**

## WORK EXPERIENCE

### Software Engineering Intern — Jun. 2026 – Aug. 2026

***Google*** — *Taipei, Taiwan*

- **Test Infrastructure:** Built **3** integration test infrastructures from a **zero** baseline (**Java**, **Python**), onboarding **400+** test suites across **200+** smart home device types, with **90%+** unit test coverage for the infrastructure itself
- **Automated Failure Triage:** Built an LLM-backed service that triages scheduled integration test runs and posts root-cause summaries directly onto issue tickets, replacing manual log inspection and cutting failure narrow-down time by **20+ minutes** per issue
- **Local Reproduction Tooling:** Extended the service to reproduce device failures locally against virtual devices instead of waiting on cloud runs, cutting manual reproduction time by **85%**; **shipped and adopted by the team**

### Backend Engineer — Dec. 2024 – Aug. 2025

***MaiAgent Co., Ltd.*** *– Award-Winning B2B GenAI Startup* — *Taipei, Taiwan*

- **API Optimization:** Optimized **140+** RESTful APIs through SQL query refactoring, connection pooling (Elasticsearch, Cohere, OpenAI), and Django caching; cut response time of 13 high-traffic APIs by **27.7%** overall and eliminated N+1 queries
- **CI/CD & Testing:** Built a **GitHub Actions** CI pipeline with pytest unit and E2E coverage, reaching **67%** code coverage from a **zero** baseline; reduced production hotfixes by **90%** initially and **50%** long term; managed database migrations in the CD pipeline
- **Open Source:** Contributed **14** merged pull requests to **LlamaIndex** (**15K+ stars**), fixing bugs and adding features across the AWS Bedrock, Claude, Elasticsearch, Cohere, OpenAI, and MCP Client integrations

### Backend Engineer Intern — Sep. 2024 – Dec. 2024

***MaiAgent Co., Ltd.*** *– Award-Winning B2B GenAI Startup* — *Taipei, Taiwan*

- **Data Pipeline:** Built an async indexing pipeline with **asyncio** and **Celery**, **3.5x faster** parsing while scaling **3M to 20M+** text chunks
- **Real-Time Communication:** Implemented a **WebSocket** notification system with **Redis** pub/sub for file parsing status; built event-driven architecture for agent state transitions, pushing render triggers to the frontend for live UI updates

## EDUCATION

### Rice University - Top 20 U.S. University — Aug. 2025 – Dec. 2026 (Expected)

M.C.S., Computer Science (GPA: 4.00/4.00) — *Houston, TX*

### National Yang Ming Chiao Tung University (NYCU) - Top 3 Taiwan University — Sep. 2020 – Jun. 2024

B.S., Industrial Engineering and Management (GPA: 4.07/4.30, Two-Time Dean's List) — *Hsinchu, Taiwan*

Minor: Computer Science (Domain GPA: 4.13/4.30)

## PROJECTS

**Relational Database Engine from Scratch** | *C++, B+ Tree, Query Optimization, TPC-H* — **Aug. 2025 – Dec. 2025**

- **Storage Engine:** Built the C++ persistence layer, a buffer manager with **LRU** eviction and page pinning, slotted-page record storage, and a **B+ tree** index supporting range scans
- **Query Processing:** Implemented the relational operators, including a **sort-merge join** over external merge sort that spills to disk, under a SQL compiler and a cost-based optimizer that orders joins from cardinality statistics; ran **TPC-H** at scale factor 1

**Full-Stack Social Network (Ricebook)** | *React, Redux Toolkit, Express, MongoDB, OAuth 2.0, Vitest* — **Sep. 2025 – Dec. 2025**

- **Full-Stack Build:** Built a **React 19** single-page app on Redux Toolkit against an **Express 5**/MongoDB API, with session-cookie auth plus **Google OAuth 2.0** linking, a follower graph, a comment-threaded feed, and Cloudinary image upload
- **Testing & Deployment:** Covered the Redux store with **30** Vitest suites at **97%** statement coverage and the API with Jasmine integration tests; deployed frontend and backend to separate hosts with CORS-aware session config

## AWARDS

**Presidential Hackathon Winner (2024, Top 5 Nationally)** - Urban noise agent: structured-output LLM pipeline and tool calling

**23rd Golden Peak Award (2025)** - Outstanding Commercial Product, MaiAgent AI Platform

## SKILLS

**Languages:** Python, Java, C/C++, TypeScript | **Backend:** Django, FastAPI, Express.js | **Frontend:** HTML, CSS, JavaScript, React
**APIs:** RESTful, WebSocket, Redis Pub/Sub | **Databases:** PostgreSQL, MongoDB, Elasticsearch, Neo4j, Milvus | **Async:** asyncio, Celery
**Testing:** pytest, JUnit, Vitest | **DevOps:** GitHub Actions, Docker, git, Nginx, Shell Script | **AWS:** EC2, S3, RDS, ElastiCache